Voren® Enteric Microencapsulated Capsule

Voren, the brand of Diclofenac Sodium, is an anti-inflammatory, analgesic and antipyretic agent. Chemically it is Sodium a - [0 – (2, 6 dichloroanilino) phenyl] acetate.

Ingredient(s):
Each capsule contains:
Diclofenac Sodium Pellets 165mg
(eq. to Diclofenac Sodium ………………...……….…………....... 50mg)

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetics):
Pharmacodynamics:
1. Diclofenac Sodium is a nonsteroidal, anti-inflammatory agent. It has a potent anti-inflammatory, analgesic and antipyretics activities. Its mode of action includes potent cyclo-oxygenase inhibition and enhances uptake of arachidonic acid into cellular triglyceride pools, resulting in a dual effect on the cyclo-oxygenase and lipoxygenase pathways.
2. Diclofenac Sodium exhibits very marked anti-rheumatic, anti-inflammatory, analgesic and antipyretics properties and is thus suitable for use in inflammatory and degenerative rheumatism disease, as well as for the treatment of rheumatoid arthritis and other rheumatic disorders.
3. Diclofenac Sodium interacts with the arachidonic acid cascade at the level of cyclo-oxygenase so that Diclofenac Sodium prevents the formation of thromboxanes, prostaglandin and prostacycline.

Pharmacokinetics:
1. Diclofenac Sodium is a non-steroid-type of anti-inflammatory agent and is widely used clinically because of its strong analgesics, antipyretics and anti-inflammatory effects.
2. Like many other NSAIDs, diclofenac is highly bound to plasma protein, mainly albumin and the degree of binding has been shown to be > 99.5%.
3. Diclofenac entered the synovial fluid quickly, since the mean 2 hours synovial concentration was approximately 2/3 that in plasma (197 and 293 $\mu g.L^{-1}$ respectively) and diclofenac persisted in the synovial fluid much longer (mean 7 hours concentration $205\mu g.L^{-1}$) compared with plasma ($52\mu g.L^{-1}$).
4. Diclofenac Sodium undergoes extensive biotransformation in the liver during the first passage through which about 50% is metabolized. Approximately 1/3 of the dose is excreted in the bile and approximately 2/3 via the kidney in the form of glucuronides.
   The mean terminal elimination half-life of Diclofenac Sodium is 1.2 to 1.8 hours and metabolite possesses about 1/30 of the anti-inflammatory activity of parent drug (Diclofenac Sodium).
5. In order to eliminate the gastrointestinal adverse effects of Diclofenac Sodium, effective enteric-coated products were developed by Yung Shin Pharmaceutical. Advantages of Diclofenac Sodium in Enteric Microencapsulated capsules includes ready distribution over a large surface area, thus minimizing the risk of local damage caused by the dumping effects of the enteric-coated tablets.
   Furthermore diclofenac enteric-microencapsulated capsules are also less dependent on gastric transit time. They may attain more constant plasma level, achieve slow release effect, give higher accuracy in reproducibility between doses, and provide less decrease in bioavailability.

Indication(s):
Inflammatory and degenerative forms of rheumatism; rheumatoid arthritis, ankylosing spondylitis, osteoarthritis and spondyl arthritis. Non-articular rheumatism, post traumatic inflammation and swelling.

Dosage and Administration:
The usual adult dose is 1 capsule, twice daily.
To be dispensed on physician's prescription.
After assessing the risk/ benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

As a general recommendation, the dose should be individually adjusted. Adverse effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section Precaution/Warning).

Established cardiovascular disease or significant cardiovascular risk factors
Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100 mg daily if treated for more than 4 weeks (see section Precaution/ Warning).

Route of Administration(s): Oral

Contraindication:
1. Diclofenac Sodium should not be used in aspirin-sensitive patients, because of the likelihood of the production of asthma rhinitis and dyspnoea in patients who exhibits these symptoms when receiving aspirin.
2. It is contraindicated in patient with active peptic ulcer as gastrointestinal bleeding has been reported.
3. It is contraindicated with in patient with severe cardiac failure (see section Precaution/Warning).

Side Effect(s) / Adverse Reaction(s):
Gastrointestinal side effects are the most frequent adverse effects of Diclofenac Sodium.
Nervous system side effects are headache, insomnia, somnolence and dizziness.
Renal toxicity, edema, skin rashes, palpitation, minor hearing disorders. Thrombocytopenia and eosinophilia may occur rarely.
Dermatological:
Occasional: rashes or skin eruptions
Cases of hair loss, bullous eruption, erythema multiforme, Stevens Johnson syndrome, toxic epidermal necrolysis (Lyell's syndrome), and photosensitivity reactions have been reported.
Cardiac disorder : Uncommon*: Myocardial infarction, cardiac failure, palpitations, chest pain.
Kounis syndrome: Frequency "not known"
*The frequency reflects data from long-term treatment with a high dose (150mg/day).

Description of selected adverse drug reactions

Arteriothrombotic events
Meta-analysis and pharmacoepidemiological data point towards an increased risk of arteriothrombotic events (for example myocardial infarction) associated with the use of diclofenac, particularly at a high dose (150mg daily) and during long-term treatment (see section Precaution/Warning).

Precaution(s) / Warning(s):
1. Caution should be exercised in patients with a history of blood dyscrasias or disorders of coagulation.
2. Fluid retention and oedema have been observed in some patients taking NSAIDs, therefore caution is advised in patients with fluid retention or heart failure.
3. The anti-inflammatory, antipyretic, and analgesic effects of Voren may mask the normal signs of infections. The physician should be alert to any development of infection in patients receiving Voren.
4. Patients with severe hepatic, cardiac or renal insufficiency or the elderly should be kept under close surveillance.
5. Use of Diclofenac in patients with hepatic porphyria may trigger an attack.
6. NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reaction and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.
7. It should not be used in conjunction with anti-inflammatory drugs or other non-steroidal

anti-inflammatory drugs including aspirin. Diclofenac Sodium in common with other NSAIDs, can reversibly inhibit platelet aggregation.

8. Safe condition for its use in pregnant women have not been established. Diclofenac Sodium is not recommended for nursing mothers because it will be excreted in the milk.
9. Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.
10. There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.
11. NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking anti-hypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.
12. Gastrointestinal effects

    All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulceration occurs in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patients about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

    NSAIDs, including diclofenac, may be associated with increased risk of gastrointestinal anastomotic leak. Close medical surveillance and caution are recommended when using diclofenac after gastrointestinal surgery.
13. Cardiovascular effects

    Treatment with NSAIDs including diclofenac, particularly at high dose and in long term, maybe associated with an increased risk of serious cardiovascular thrombotic events (including myocardial infraction and stroke).

    Treatment with diclofenac is generally not recommended in patients with established cardiovascular diseases (congestive heart failure, established ischemic heart diseases, peripheral arterial diseases) or uncontrolled hypertension. If needed, patients with established cardiovascular diseases, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes melilitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100mg daily when treatment continues for more than 4 weeks.

    As the cardiovascular risks of diclofenac may increase with dose and duration of exposure, the lowest effective daily dose should be used for the shortest duration possible. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially when treatment continues for more than 4 weeks.

    Patients should remain alert for the signs and symptoms of serious arteriothrombotic events (e.g. chest pain, shortness of breath, weakness, slurring of speech), which can occur without warnings. Patients should be instructed to see a physician immediately in case of such an event.

WARNINGS

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (eg. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (eg. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Pregnancy and Lactation:
Use in pregnant women has not been established. Nor is Diclofenac Sodium recommended for use in nursing mothers because it will be excreted in the milk.

Drug Interaction(s):
This drug has an affinity for serum albumin, and may displace other drugs which are also bound to albumin. Concomitant treatment with Potassium sparing diuretics may be associated with increased serum potassium levels. Concomitant administration with aspirin is not recommended because Voren is displaced from binding site and resulting in lower plasma concentrations and peak plasma level. Concurrent therapy with Voren and Warfarin requires a close monitoring of patients to be certain that no change in their anticoagulant dosage is required. Patients with altered renal function should be observed for the development of the specific toxicities of concomitant administration of Voren and Digoxin, Methotrexate and Cyclosporine. Patients receiving oral hypoglycemics should be observed for signs of toxicity of these drugs.

Symptoms and Treatment for Overdosage, and Antidote(s):
Symptoms of overdose reported have generally reflected the gastrointestinal, renal and CNS toxicities of this medication. More serious overdosage effects such as gastrointestinal haemorrhage, acute renal failure, convulsion and coma have been reported.
Should accidental overdosage occur, the stomach should be emptied by inducing emesis or by careful gastric lavage followed by the administration of activated charcoal. Vital functions should be monitored and supported. Because it is firmly bound to plasma proteins, hemodialysis and peritoneal dialysis may be of little value.

Shelf-Life:
2 years from the date of manufacture.

Storage Condition(s):
Store at temperature below 30°C. Protect from light and moisture.

Product Description(s) and Packing(s):
A size #4 red cap and transparent body hard gelatin capsule containing milky-white pellets.
Blister packing of 10's x 10.

Y. S. P.®

Manufacturer and Product Registration Holder:
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